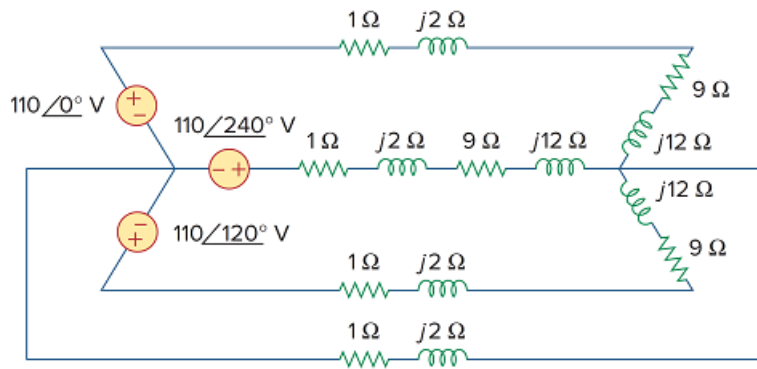


Problem

Given the circuit in Fig. 12.57 below, find the total complex power absorbed by the load.

Figure 12.57



Step-by-step solution

Step 1 of 1

As a balanced three phase system, we can use the per phase equivalent shown below.

$$I_a = \frac{110\angle 0^\circ}{(1+j2)+(9+j12)} = \frac{110\angle 0^\circ}{10+j14}$$

$$S_p = \frac{1}{2} |I_a|^2 Z_y = \frac{1}{2} \frac{(110)^2}{(10^2+14^2)} (9+j12)$$

The complex power is,

$$S = 3S_p = \frac{3(110)^2}{2 \cdot 296} (9+j12)$$

$$S = 551.86 + j735.81 \text{ VA} .$$